Krishnaa **Vadivel** | Electrical Engineer | Semiconductor Devices | Photonics | FPGA Systems

516-853-5663 • srikrishnaa.careers@gmail.com • sri-krishnaa-ece-phd
Website | GitHub: krishnaa423 | LeetCode: krishnaa42342 | Docker Hub: krishnaa42342

Summary

Electrical engineering PhD researcher with Cornell and Yale ECE training across semiconductor physics, embedded systems, photonics, and accelerator-enabled scientific computing. Experience spans FPGA and MCU development, CUDA-based engineering tools, semiconductor-device modeling, and digital-design projects in Verilog.

Experience

Yale University, Electrical Engineering
PhD Research Assistant / Embedded Systems TA *2021–present*

- Conduct research at the intersection of semiconductor physics, condensed-matter modeling, and high-performance scientific computing, with supporting coursework and theory development in semiconductor devices, photonics, and signals/systems.
- Teach and support embedded systems instruction, reinforcing firmware, instrumentation, debugging, and system-integration skills relevant to practical EE development.
- Build numerical and software workflows for device-scale and materials-scale simulation using Python, C++, HPC toolchains, and accelerator-enabled platforms.
- Co-authored a 2025 *Journal of the American Chemical Society* paper and delivered APS presentations in 2024, 2025, and 2026 on self-trapped exciton formation and related excited-state materials physics.

Probe Technologies Inc. / Weatherford Wireline Services
Software and Embedded Systems Engineer *2019–2021*

- Developed embedded firmware and FPGA-based sensor-logging functionality for downhole geological tools using Lattice FPGA, PIC32, dsPIC, STM32, and C8051-class platforms.
- Built CUDA-based software for large acoustic waveform processing, engineering visualization, and diagnostic analysis in field-tool workflows.
- Supported circuit and system bring-up with oscilloscope-based validation, field testing, and targeted PCB updates in Altium.

FindLight
Photonics Technical Writer *2018–2019*

- Wrote technical content on lasers, integrated optics, and optoelectronic components for engineering-facing audiences.

Selected Projects

Biosignal Robotic Hand: Published Cornell ECE project combining surface EMG sensing, analog filtering, PIC32 control logic, and servo actuation for robotic finger motion.

FPGA AI Accelerator: Small Verilog matrix-multiply accelerator with simulation-backed verification for ML-oriented datapath design.

Semiconductor Simulation: Numerical modeling of semiconductor electrostatics, drift-diffusion transport, and pn-junction-style device behavior.

Integrated Photonics Modeling: FDTD simulation of resonator and band-pass style photonic structures using Meep.

Digital Design: Single-cycle MIPS processor in Verilog with self-checking test bench and documentation.

Education

Yale University
PhD, Electrical Engineering *2021–present*

Yale University
MPhil, Electrical Engineering *2023*

Yale University
MS, Electrical Engineering *2023*

Cornell University
MEng, Electrical and Computer Engineering *2017–2018*

Cornell University
BS, Electrical and Computer Engineering *2013–2017*

Technical Skills

Hardware: Verilog, FPGA, MCU firmware, PCB bring-up, instrumentation, oscilloscope-based debugging

Devices: Semiconductor physics, pn junctions, MOSFET concepts, transport modeling, thin films

Photonics: Integrated photonics, lasers, optics, FDTD simulation

Software: Python, C, C++, CUDA, Linux

Systems: Docker, Kubernetes, scientific computing, HPC workflows